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414. On Polyzomal Curves (continued)

ANNEX III. ON THE NORM OF

$$(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma},$$

WHEN THE CENTRES ARE IN A LINE.

The norm of $\sqrt{U}+\sqrt{V}+\sqrt{W}$ is

$$(1,1,1,-1,-1,-1 \mid U,V,W),$$

whence that of $\sqrt{U}+\sqrt{U'}+\sqrt{V}+\sqrt{V'}+\sqrt{W}+\sqrt{W'}$ is

$$\begin{aligned}&(1,1,1,-1,-1,-1 \mid U,V,W)\\ &\qquad+(1,1,1,-1,-1,-1 \mid U',V',W')\\ &\qquad\qquad+2(1,1,1,-1,-1,-1 \mid U,V,W \,|\, U',V',W'),\end{aligned}$$

where the last term is $=2$ into

$$U'(-U-V+W)+V'(-U+V-W)+W'(-U-V+W);$$

and the norm of $\sqrt{U+U'+U''}+\sqrt{V+V'+V''}+\sqrt{W+W'+W''}$ is obviously composed in a similar manner.

Now, applying the formula to obtain the norm of

$$(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma},$$

the expression contains six terms, two of which are at once seen to vanish; and writing for shortness ( ) in place of $(1,1,1,-1,-1,-1)$ the remaining terms are

$$\begin{aligned}&(\ )\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma\}\\ &\qquad+2(\ )\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma \,|\, (b-c)^2\alpha',(c-a)^2\beta',(a-b)^2\gamma'\}\\ &\qquad+2\theta(\ )\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma \,|\, (b-c)^2,(c-a)^2,(a-b)^2\}\\ &\qquad\qquad+2\theta(\ )\{(b-c)^2\alpha',(c-a)^2\beta',(a-b)^2\gamma' \,|\, (b-c)^2,(c-a)^2,(a-b)^2\}\end{aligned}$$

the first of these terms requires no reduction; the second, omitting the factor 2, is

$$\begin{aligned}&(b-c)^2\alpha\,[(b-c)^2\alpha'-(c-a)^2\beta'-(a-b)^2\gamma']\\ &\qquad+(c-a)^2\beta\,[-(b-c)^2\alpha'+(c-a)^2\beta'-(a-b)^2\gamma']\\ &\qquad\qquad+(a-b)^2\gamma\,[-(b-c)^2\alpha'-(c-a)^2\beta'+(a-b)^2\gamma'],\end{aligned}$$

which is

$$=2(a-b)(b-c)(c-a)\,[bc(b-c)\alpha+ca(c-a)\beta+ab(a-b)\gamma].$$

Similarly the third term, omitting the factor 2θ, is

$$\begin{aligned}&(b-c)^2\alpha\,[(b-c)^2-(c-a)^2-(a-b)^2]\\ &\qquad+(c-a)^2\beta\,[-(b-c)^2+(c-a)^2-(a-b)^2]\\ &\qquad\qquad+(a-b)^2\gamma\,[-(b-c)^2-(c-a)^2+(a-b)^2],\end{aligned}$$

which is

$$=2(a-b)(b-c)(c-a)\,[(b-c)\alpha+(c-a)\beta+(a-b)\gamma],$$

and for the last term, omitting the factor 2θ, this may be deduced therefrom by writing (α',β',γ') in place of (α,β,γ), viz., it is

$$=-2(a-b)^2(b-c)^2(c-a)^2.$$

Hence, restoring the omitted factors, and collecting, we find

$$\begin{aligned}
&\mathrm{Norm}\{(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma}\}\\
&= (b-c)^4\alpha^2+(c-a)^4\beta^2+(a-b)^4\gamma^2-2(c-a)^2(a-b)^2\beta\gamma-2(a-b)^2(b-c)^2\gamma\alpha-2(b-c)^2(c-a)^2\alpha\beta\\
&\qquad + 4\theta(a-b)(b-c)(c-a)\,[(b-c)\alpha+(c-a)\beta+(a-b)\gamma]\\
&\qquad + 4(a-b)(b-c)(c-a)\,[bc(b-c)\alpha+ca(c-a)\beta+ab(a-b)\gamma]\\
&\qquad - 4\theta^2(a-b)^2(b-c)^2(c-a)^2.
\end{aligned}$$

Hence, first writing $a-x$, $b-x$, $c-x$ in place of a, b, c; then y^2 for θ, and $(-\alpha'', -\beta'', -\gamma'')$ for (α, β, γ); and finally introducing z for homogeneity, we find

$$\begin{gathered}
\text{Norm}\{(b-c)\sqrt{(x-az)^2+\alpha'' z^2}+(c-a)\sqrt{(x-bz)^2+\beta''^2 z^2}+(a-b)\sqrt{(x-cz)^2+\gamma''^2 z^2}\} = z^8 \text{ into} \\
z^{-4}\{(b-c)^4\alpha''^2+(c-a)^4\beta''^2+(a-b)^4\gamma''^2 \\
-2(c-a)^2(a-b)^2\beta''\gamma''-2(a-b)^2(b-c)^2\gamma''\alpha''-2(b-c)^2(c-a)^2\alpha''\beta''\} \\
-4y^2(b-c)(c-a)(a-b)\,[(b-c)\alpha''+(c-a)\beta''+(a-b)\gamma''] \\
-4(b-c)(c-a)(a-b)\{(b-c)\alpha''[2bc-2x(b+c)+x^2] \\
+(c-a)\beta''[2ca-2x(c+a)+x^2]+(a-b)\gamma''[2ab-2x(a+b)+x^2]\} \\
-4y^4(b-c)^2(c-a)^2(a-b)^2,
\end{gathered}$$

so that the equation $(b-c)\sqrt{S}+(c-a)\sqrt{S'}+(a-b)\sqrt{S''}=0$, in its rationalised form, contains $(z^4=0)$ the line infinity twice, and the curve is thus a conic. If $\alpha''=\beta''=\gamma''=k''^2$, then the expression of the norm is

$$= z^4 \text{ into } -4(a-b)^2(b-c)^2(c-a)^2(y^4-k''^4 z^4),$$

viz., when the three circles have each of them the same radius k'', the curve is the pair of parallel lines $y^4-k''^4z^4=0$; and in particular when $k''=0$, or the circles reduce themselves each to a point, then the curve is $y^4=0$, the axis twice.

Annex IV. On the Trizomal Curves

$$\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0, \text{ which have a Cusp, or two Nodes.}$$

The trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, has not in general any nodes or cusps: in the particular case where the zomal curves are circles, we have however seen how the ratios $l:m:n$ may be determined so that the curve shall acquire a node, two nodes, or a cusp; viz., regarding a, b, c as current areal coordinates, we have here a conic $\alpha+\beta+\gamma=0$, the locus of the centres of the variable circle, and the solution depends on establishing a relation between this conic and the orthotomic circle or Jacobian of the three given circles. I have in my paper "Investigations in connection with Casey's Equation," *Quart. Math. Jour.* vol. viii. (1867), pp. 334—342, [395] given, after Professor Cremona, a solution of the general question to find the number of the curves $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, which have a cusp, or which have two nodes, and I will here reproduce the leading points of the investigation. I remark, that although one of the loci involved in it is the same as that occurring in the case of the three circles (viz., we have in each case the Jacobian of the given curves), the other two loci Σ and Λ, which present themselves, seem to have no relation to the conic of centres which is made use of in the particular case.

We have the curves $U = 0$, $V = 0$, $W = 0$, each of the same order r; and considering a point the coordinates whereof are (l, m, n), we regard as corresponding to this point the curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, say for shortness, the curve Ω, being as above a curve of the order $2r$, having r^2 contacts with each of the given curves $U = 0$, $V = 0$, $W = 0$. As long as the point (l, m, n) is arbitrary, the curve Ω has not any node, and in order that this curve may have a node, it is necessary that the point (l, m, n) shall lie on a certain curve Λ; this being so, the node will, it is easy to see, lie on the curve J, the Jacobian of the three given curves; and the curves J and Λ will correspond to each other point to point, viz., taking for (l, m, n) any point whatever on the curve Λ, the curve Ω will have a node at some one point of J; and conversely, in order that the curve Ω may be a curve having a node at a given point of J, the point (l, m, n) must be at some one point of the curve Λ.

The curve Λ has, however, nodes and cusps; each node of Λ corresponds to two points of J, viz., for (l, m, n) at a node of Λ, the curve Ω is a binodal curve having a node at each of the corresponding points of J; each cusp of Λ corresponds to two coincident points of J, viz. for (l, m, n) at a cusp of Λ, the curve Ω has a node at the corresponding point of J. The number of the binodal curves Ω is thus equal to the number of the nodes of Λ, and the number of the cuspidal curves Ω is equal to the number of the cusps of Λ; and the question is to find the Plückerian numbers of the curve Λ.

This Professor Cremona accomplished in a very ingenious manner, by bringing the curve Λ into connexion with another curve Σ (viz., Σ is the locus of the nodes of those curves $lU + mV + nW = 0$ which have a node), and the result arrived at is that for the curve Λ

$$\begin{aligned}
\text{Order} &= 3(r-1)(3r-2),\\
\text{Class} &= 6(r-1)^2,\\
\text{Nodes} &= \tfrac{1}{2}(r-1)(27r^3 - 63r^2 + 22r + 16),\\
\text{Cusps} &= 3(r-1)(7r-8),\\
\text{Double tangents} &= \tfrac{1}{2}(r-1)(12r^3 - 36r^2 + 19r + 16),\\
\text{Inflexions} &= 12(r-1)(r-2);
\end{aligned}$$

so that, finally, the number of the cuspidal curves $\sqrt{lU}+\sqrt{mV}+\sqrt{nW} = 0$, is found to be $= 3(r-1)(7r-8)$, and the number of the binodal curves of the same form is found to be $= \frac{1}{2}(r-1)(27r^3-63r^2+22r+16)$. When the given curves are conics, or for $r = 2$, these numbers are $= 18$ and 36 respectively; but the formulae are not applicable to the case where the conics have a point or points of intersection in common; nor, consequently, to the case of the three circles.

415. Corrections and Additions to the Memoir on the Theory of Reciprocal Surfaces (Phil. Trans. vol. clix. 1869, [411]).

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII. (for the year 1872), pp. 83—87. Received July 22,—Read November 16, 1871.]

1. I am indebted to Dr Zeuthen for the remark that although the "off-points" and "off-planes," as explained in the memoir, are real singularities, they are not the singularities to which the θ, θ' of the formulae refer. The most convenient way of correcting this is to retain all the formulae with θ, θ' as they stand, but to write ω, ω' for the number of "off-points" and "off-planes" respectively; viz. we thus have

ω, off-points,
θ, unexplained singular points,

and

ω', off-planes,
θ', unexplained singular planes,

the formulae as they stand, taking account of the unexplained singularities θ and θ', but not taking any account at all of the off-points and off-planes ω, ω'. The extended formulae in which these are taken into account are:

$$a(n-2) = \kappa - \beta + \rho + 2\sigma + 3\omega, \tag{1}$$
$$b(n-2) = \rho + 2\beta + 3\gamma + 3i, \tag{2}$$
$$c(n-2) = 2\sigma + 4\beta + \gamma + i + \omega, \tag{3}$$

$$a(n-2)(n-3) = 2(\delta - C - \beta\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2p - j), \tag{4}$$
$$b(n-2)(n-3) = 4k + (ab - 2p - j) + 3(bc - 3\beta - 2\gamma - i), \tag{5}$$
$$c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i), \tag{6}$$

which replace Salmon's original formulae (A) and (B).

2. In the reciprocal formulae (A′) and (B′), for ω, ω' we must of course write ω', ω respectively: viz. these are

$$a'(n'-2) = \kappa' - B + \rho' + 2\sigma' + 3\omega', \tag{7}$$
$$b'(n'-2) = \rho' + 2\beta' + 3\gamma' + 3i', \tag{8}$$
$$c'(n'-2) = 2\sigma' + 4\beta' + \gamma' + i' + \omega', \tag{9}$$

$$a'(n'-2)(n'-3) = 2(B - C - 3\omega') + 3(a'c' - 3\sigma' - \chi' - 3\omega') + 2(a'b' - 2p' - j'), \tag{10}$$

$$b'(n'-2)(n'-3) = 4k' + (a'b' - 2p' - j') + 3(b'c' - 3\beta' - 2\gamma' - i'), \tag{11}$$
$$c'(n'-2)(n'-3) = 6h' + (a'c' - 3\sigma' - \chi' - 3\omega') + 2(b'c' - 3\beta' - 2\gamma' - i'). \tag{12}$$

The equations involving the points χ, ω, and the planes χ', ω' respectively are

$$c(-5n+6) + 3\omega = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega, \tag{13}$$
$$c'(5n-12) - 3\omega' = 6r' + 18\beta' + 5\gamma' + 4i' + 2\chi' + 3\omega', \tag{14}$$

(compare Salmon, p. 586, equations (C) and (D) near the bottom). The first of these is obtained from the value of $c(n-2)(n-3)$ by means of the equation

$$c' - c = 2h + 3\beta \quad \text{or} \quad c' = c + 2h + 3\beta,$$

which gives

$$(c + 2h + 3\beta)(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i);$$

substituting herein for ac and bc their values and reducing, we find

$$c\{(n-2)(n-3) - a - 2b\} = 6h - 6\beta - 6\sigma - 2\chi - 3\omega - 4\gamma - 2i,$$

where

$$(n-2)(n-3) - a - 2b = -5n + 6,$$

and the equation thus becomes

$$c(-5n+6) = 6h - 6\beta - 6\sigma - 2\chi - 3\omega - 4\gamma - 2i,$$

or, since $6h - 6\sigma = 6r$,

$$c(-5n+6) = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega.$$

The reciprocal of the first of these is

$$c'(-5n'+6) = 6h' - 6\beta' - 6\sigma' - 2\chi' - 3\omega' - 4\gamma' - 2i',$$

viz. writing $a = n(n-1) - 2b - 3c$, and $\kappa = 3n(n-2) - 6b - 8c$, this is

$$= 4n(n-2) - 8b - 11c - 2j - 3\chi' - 2C - 4B - 3\omega';$$

and it thus appears that the order c' of the spinode curve is reduced by 3 for each off-plane ω'.

4. As to the other two equations, writing for ρ, σ their values, these become

$$j + 6t + 3i + 5\beta + 6\gamma = b(2n-4) - 2q,$$
$$2\chi + 3\omega + 4i + 18\beta + 6\gamma = c(5n-12) - 6r + 3\omega,$$

equations which admit of a geometrical interpretation. In fact, when there is only a nodal curve, the first equation is

$$j + 6t = b(2n-4) - 2q,$$

which we may verify when the nodal curve is a complete intersection, $P=0$, $Q=0$; for if the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, where the degrees of A, B, C, P, Q are respectively $n-2f$, $n-f-g$, $n-2g$, f, $g=0$ (so that $b=fg$), then the number of pinch-points is thus $= fg(2n-2f-2g)$, $= (2n-4)fg - 2fg(f+g-2)$; but for the curve $P=0$, $Q=0$ we have $t=0$, and its order and class are $b = fg$, $q = fg(f+g-2)$, or the formula is thus verified.

Similarly, when there is only a cuspidal curve, the second equation is

$$2\chi + 3\omega = c(5n-12) - 6r + 3\omega,$$

which equation may be verified when the cuspidal curve is a complete intersection, $P=0$, $Q=0$, and the points χ, ω are given as the intersections of the curve with the surface $(A, B, C \mid N, -M)^2 = 0$.

Now $AC - B^2$ vanishing for $P=0$, $Q=0$ we must have $A = \Lambda\alpha^2 + A'$, $B = \Lambda\alpha\beta + B'$, $C = \Lambda\beta^2 + C'$, where A', B', C' vanish for $P=0$, $Q=0$; and thence $M = \Lambda M' + M''$, $N = \Lambda N' + N''$, where M'', N'' vanish for $P=0$, $Q=0$. The equation $(A, B, C \mid N, -M)^2 = 0$, writing therein $P=0$, $Q=0$, thus becomes $\Lambda'(N'\alpha - M'\beta)^2 = 0$; and its intersections with the curve $P=0$, $Q=0$ are the points $P=0$, $Q=0$, Λ' each three times, and the points $P=0$, $Q=0$, $N'\alpha - M'\beta = 0$ each twice; viz. they are the points $2\chi + 3\omega$. But if the degree of Λ is $=\lambda$, then the degrees of N', M', α, β are $2n-3f-2g-\lambda$, $2n-2f-3g-\lambda$, $n-2f-\lambda$, $n-2g-\lambda$, whence the degree of $\Lambda'(N'\alpha - M'\beta)^2$ is $= 5n-6f-6g$, and the number of points is $= fg(5n-6f-6g)$, viz. this is $= fg(5n-12) - 6fg(f+g-2)$, or it is $= c(5n-12) - 6r$; so that β being $=0$, the equation is verified.

5. It was also pointed out to me by Dr Zeuthen that in the value of $24t$ given in No. 10 the term involving χ should be -6χ instead of $+6\chi$, and that in consequence the coefficients of β' are erroneous in several others of the formulae. Correcting these, and at the same time introducing the terms in ω, ω', and writing down also the terms in θ' as they stand, we have

$$\begin{aligned}
4\rho &= \ldots - 2\chi + 3\theta - 3\omega,\\
24t &= \ldots - 6\chi + 9\theta - 9\omega,\\
2\sigma &= \ldots - \theta - \omega,\\
8\beta &= \ldots + 6\chi - 9\theta + 9\omega,\\
8k &= \ldots - 6\chi + 17\theta - 25\omega,\\
2j &= \ldots + 6\chi - 9\theta + 15\omega,\\
8\omega' &= \ldots - 30\chi + 21\theta - 45\omega,\\
c' &= \ldots - 12\chi + 10\theta - 20\omega.
\end{aligned}$$

The equations of No. 11, used afterwards, No. 53, should thus be

$$\begin{aligned}
&4i + 6r = (5n-12)c - 18\beta - 5\gamma - 2\chi + 3\theta - 3\omega,\\
&\qquad - 24t - 8\beta + 18r = (-8n+16)b + (15n-36)c - 34\beta + 6\gamma + 4j - 6\chi + 9\theta - 9\omega;
\end{aligned}$$

and from these I deduce

$$44q' + 6r = (44n - 88)b + (-45n + 63)c - 4\tfrac{2}{3}\beta - 3\tfrac{1}{3}\gamma - 13i - 8j - 22\chi + \ldots$$

6. In No. 32 we have (without alteration) $\theta = 16$; but in the application (Nos. 40 and 41) to the surface $FP^2 + GR - Q^2 = 0$ we have $b = 0$, and there are $t = fpq$ off-points, $F = 0$, $P = 0$, $Q = 0$, and $\chi = 9Pl$ close-points, $G = 0$, $P = 0$, $Q = 0$. The new equations involving ω are thus satisfied.

7. I have ascertained that the value of β' obtained, Nos. 51 to 64 of the memoir, is inconsistent with that obtained in the "Addition" by consideration of the deficiency, and that it is in fact incorrect. The reason is that, although, as stated No. 53, the values of two of the coefficients D, E may be assumed at pleasure, they cannot, in conjunction with a given system of values of A, B, C, be thus assumed at pleasure; viz. A, B, C being $= 110$, 272, 44 respectively, the values of D, E are really determinate. I have no direct investigation, but by working back from the formula in the Addition I find that we must have $D = 271$, $E = 315$; the values of the remaining coefficients then are

$$F = \tfrac{7}{2},\quad G = -\tfrac{7}{2},\quad H = -\tfrac{15}{2},\quad I = -198;$$

or the formula is

$$\begin{aligned}
\beta' = 2n(n-2)(11n-24) &- (110n-272)b + 44j\\
&+ \tfrac{1}{2}(271n-315)c + \tfrac{7}{2}i - \tfrac{7}{2}\beta - 4\tfrac{1}{2}\gamma - 198\,\epsilon\\
&- hC - gB - i\,\iota - \chi\,j - \theta\,\beta - \ldots\omega\\
&+ h'C + g'B + i'\,\iota' + \chi'\,j' + \theta'\,\beta' - f'\,\omega';
\end{aligned}$$

but I have not as yet any means of determining the coefficients f, f' of the terms in ω, ω'.

From the several cases of a cubic surface we obtain as in the memoir; but applying to the same surfaces the reciprocal equation for β, instead of the results of the memoir, we find

$$\begin{aligned} h &= \ \ 24, & h' &= \ \ 4,\\ g+2\chi &= \ \ 45, & g'+2\chi' &= -198,\\ g+g' &= \ \ 9, && \\ \theta+\theta' &= 518, && \\ \lambda+\lambda' &= -2, && \end{aligned}$$

(so that now $\lambda+\lambda'=-2$, as is also given by the cubic scroll). And combining the two sets of results, we have

$$\begin{aligned} h &= 24, & h' &= \ \ 4,\\ g &= 45-2\chi, & g' &= -198-2\chi',\\ \theta &= \ \ 9, && \\ \lambda &= -1+\tfrac{1}{2}\theta, && \end{aligned}$$

but the coefficients g, χ, χ', f, f' are still undetermined. To make the result agree with that of the Addition, I assume $\chi=-86$, $\chi'=-1$, $\theta=+28$; whence we have

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) &- (110n-272)b+44j\\ &- (271n-315)c+\tfrac{7}{2}i-\tfrac{7}{2}\beta+198\,\epsilon\\ &- 24C-28B+86\,\iota-\omega-24\theta+\ldots\omega\\ &+ 4C'+10B'+\iota'+\chi'\,j'+8\chi'-4\omega'; \end{aligned}$$

and if we substitute herein the foregoing value of $44q'+6r$, we obtain

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) &+ (-66n+184)b+(-93n+252)c\\ &+ 15\tfrac{1}{3}\beta+9\tfrac{2}{3}\gamma+6\iota-24\theta-28B-\iota-2\tfrac{1}{3}j-3\tfrac{1}{3}\chi+\theta-f\,\omega\\ &+ 4C'+10B'+\iota'+\chi'\,j'+8\chi'-4\omega', \end{aligned}$$

which, except as to the terms in ω, ω', the coefficients of which are not determined, agrees with the value given in the Addition.

Dr Zeuthen considers that in general $i'=i$; I presume this is so, but have not verified it.

416. On the Theory of Reciprocal Surfaces.

[Published as an Addition by Prof. Cayley in Dr Salmon's *Treatise on the Analytic Geometry of Three Dimensions*, 4th Ed. (8vo. Dublin, 1882), pp. 592—604.]

620. In further developing the theory of reciprocal surfaces it has been found necessary to take account of other singularities, some of which are as yet only imperfectly understood. It will be convenient to give the following complete list of the quantities which present themselves:

n, order of the surface.
a, order of the tangent cone drawn from any point to the surface.
B, number of nodal edges of the cone.
K, number of its cuspidal edges.
ρ, class of nodal torse.
σ, class of cuspidal torse.
b, order of nodal curve.
k, number of its apparent double points.
f, number of its actual double points.
t, number of its triple points.
j, number of its pinch-points.
q, its class.
c, order of cuspidal curve.
h, number of its apparent double points.

θ, number of its points of an unexplained singularity.
χ, number of its close-points.
ω, number of its off-points.
r, its class.
β, number of intersections of nodal and cuspidal curves, stationary points on cuspidal curve.
γ, number of intersections, stationary points on nodal curve.
i, number of intersections, not stationary points on either curve.
C, number of cnicnodes of surface.
B, number of binodes.

And corresponding reciprocally to these:

n', class of surface.
a', class of section by arbitrary plane.
δ', number of double tangents of section.
κ', number of its inflexions.
ρ', order of node-couple curve.
σ', order of spinode curve.
b', class of node-couple torse.
k', number of its apparent double planes.
f', number of its actual double planes.
t', number of its triple planes.
j', number of its pinch-planes.
q', its order.
c', class of spinode torse.
h', number of its apparent double planes.
θ', number of its planes of a certain unexplained singularity.
χ', number of its close-planes.
ω', number of its off-planes.
r', its order.
β', number of common planes of node-couple and spinode torse, stationary planes of spinode torse.
γ', number of common planes, stationary planes of node-couple torse.
i', number of common planes, not stationary planes of either torse.
C', number of cnictropes of surface.
B', number of its bitropes.

In all 46 quantities.

621. In part explanation, observe that the definitions of ρ and σ agree with those given, Art. 609: the nodal torse is the torse enveloped by the tangent planes along the nodal curve; if the nodal curve meets the curve of contact α, then a tangent plane of the nodal torse passes through the arbitrary point, that is, ρ will be the number of these planes which pass through the arbitrary point, viz. the class of the torse. So also the cuspidal torse is the torse enveloped by the tangent planes along the cuspidal curve; and σ will be the number of these tangent planes which pass through the arbitrary point, viz. it will be the class of the torse.

Again, as regards ρ' and σ': the node-couple torse is the envelope of the bitangent planes of the surface, and the node-couple curve is the locus of the points of contact of these planes; similarly, the spinode torse is the envelope of the parabolic planes of the surface, and the spinode curve is the locus of the points of contact of these planes.

622. The several singularities may arise as follows:

Nodal curve.—The surface may meet a plane in a curve having a node; the locus of the node is the nodal curve; each point of this curve is a point on two sheets of the surface. The pinch-points are the points of the nodal curve, each of which is also a point on the cuspidal curve; they are points where the two tangent planes of the nodal curve coincide with each other and with the tangent plane of the cuspidal curve.

Cuspidal curve.—The surface may meet a plane in a curve having a cusp; the locus of the cusp is the cuspidal curve; each point of this curve is a point on two consecutive sheets of the surface.

The several quantities b, k, f, t, j; c, h, θ, χ, ω; β, γ, i refer to the nodal and cuspidal curves respectively and to their intersections; the significations of q and r will appear presently.

623. The section by any plane is a curve; a line in the plane meets this curve in n points, and the envelope of the line is the tangential of the curve. When the plane passes through a point on the nodal curve, two of the n points unite together at the point on the nodal curve; the line has thus a double point and the tangential has a corresponding cusp. When the plane passes through a pinch-point, the two branches of the nodal curve have the same tangent, and the tangential acquires a certain singularity, say the line has there two double points; this is not a point on the cuspidal curve. The several points of the nodal curve give rise to the same singularity of the tangential, and the class of the tangential (which is the class of section a') is thus reduced; the reduction is $2q$ and we have

$$q = b' - b - 2k - 2f - 3j - 6t, \quad \text{say } q = b' - b - 2k - 2f - 3j - 6t.$$

624. So when the plane passes through a point on the cuspidal curve, two of the n points unite together at the point on the cuspidal curve; the line has thus a cusp and the tangential has a corresponding node. The several points of the cuspidal curve give rise to the same singularity of the tangential, and the class of the tangential is thus further reduced; the reduction is $3r$ and we have

$$r = c' - c - 2h - 3\beta.$$

625. At a point i the nodal curve crosses the cuspidal curve, being on the side away from the two half-sheets of the surface acnodal, and on the side of the two half-sheets crunodal, viz. the two half-sheets intersect each other along this portion of the nodal curve. There is at the point a single tangent plane, which is a plane i'; and we thus have $i = i'$.

626. As already mentioned, a cnicnode C is a point where, instead of a tangent plane, we have a tangent quadricone; and at a binode B the quadricone degenerates into a pair of planes. A cnictrope C' is a plane touching the surface along a conic; in the case of a bitrope B', the conic degenerates into a flat conic or pair of points.

627. In the original formulae for $a(n-2)$, $b(n-2)$, $c(n-2)$, we have to write $\kappa - B$ instead of κ, and the formulae are further modified by reason of the singularities θ and ω. So in the original formulae for $a(n-2)(n-3)$, $b(n-2)(n-3)$, $c(n-2)(n-3)$, we have instead of δ to write $B - C - 3\omega$; and to substitute new expressions for $[ab]$, $[ac]$, $[bc]$, viz. these are

$$[ab] = ab - 2\rho - j,$$
$$[ac] = ac - 3\sigma - \chi - \omega,$$
$$[bc] = bc - 3\beta - 2\gamma - i.$$

The whole series of equations thus is:

$$a' = n(n-1) - 2b - 3c, \tag{15}$$
$$\kappa' = 3n(n-2) - 6b - 8c, \tag{16}$$
$$\delta' = \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{9}{2}c(c-1). \tag{17}$$

$$a(n-2) = \kappa - B + \rho + 2\sigma + 3\omega, \tag{7}$$
$$b(n-2) = \rho + 2\beta + 3\gamma + 3i, \tag{8}$$
$$c(n-2) = 2\sigma + 4\beta + \gamma + i + \omega, \tag{9}$$

$$a(n-2)(n-3) = 2(\delta - C - 3\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2\rho - j), \tag{10}$$
$$b(n-2)(n-3) = 4k + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i), \tag{11}$$
$$c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i), \tag{12}$$

$$q = b' - b - 2k - 2f - 3j - 6t, \tag{13}$$
$$r = c' - c - 2h - 3\beta. \tag{14}$$

Also, reciprocal to these:

$$a' = n'(n'-1) - 2b' - 3c', \tag{15}$$
$$\kappa' = 3n'(n'-2) - 6b' - 8c', \tag{16}$$
$$\delta' = \tfrac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \tfrac{9}{2}c'(c'-1). \tag{17}$$

$$a'(n'-2) = \kappa' - B + \rho' + 2\sigma' + 3\omega', \tag{18}$$
$$b'(n'-2) = \rho' + 2\beta' + 3\gamma' + 3i', \tag{19}$$
$$c'(n'-2) = 2\sigma' + 4\beta' + \gamma' + i' + \omega', \tag{20}$$

$$a'(n'-2)(n'-3) = 2(B'-C'-3\omega') + 3(a'c'-3\sigma'-\chi'-3\omega') + 2(a'b'-2\rho'-j'), \tag{21}$$

$$b'(n'-2)(n'-3) = 4k' + (a'b'-2\rho'-j') + 3(b'c'-3\beta'-2\gamma'-i'), \tag{22}$$
$$c'(n'-2)(n'-3) = 6h' + (a'c'-3\sigma'-\chi'-3\omega') + 2(b'c'-3\beta'-2\gamma'-i'), \tag{23}$$

$$q' = b' - b - 2k' - 2f' - 3j' - 6t', \tag{24}$$
$$r' = c' - c' - 2h' - 3\beta'. \tag{25}$$

together with one other independent relation, in all 26 relations between the 46 quantities.

628. The new relation may be presented under several different forms, equivalent to each other in virtue of the foregoing 25 relations; these are

$$2(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6q + 6r + 24i + 42\beta + 30\gamma - \chi$$
$$= 2(n'-1)(n'-2)(n'-3) - 12(n'-3)(b'+c') + 6q' + 6r' + 24i' + 42\beta' + 30\gamma' - \chi', \tag{26}$$

$$26n - 12c - 4C - 10\beta + \beta - 7j - 8\chi + 16i - 4\omega$$
$$= 26n' - 12c' - 4C' - 10\beta' + \beta' - 7j' - 8\chi' + 16i' - 4\omega', \tag{27}$$

in each of which two equations Σ is used to denote the same function of the accented letters that the left-hand side is of the unaccented letters.

$$\beta' + \beta = 2n(n-2)(11n-24) + (-66n+184)b + (-93n+252)c$$
$$+ 22(2\beta+3\gamma+3i) + 27(4\beta+\gamma+i)$$
$$+ i + \beta - 24C - 28B - \iota - 2\tfrac{1}{3}j - 3\tfrac{1}{3}\chi + \theta\beta - f\omega$$
$$+ 4C' + 10B' + \iota' + 7j' + 8\chi' - 4\omega'. \tag{28}$$

Or, reciprocally,

$$\beta' + \beta = 2n'(n'-2)(11n'-24) + (-66n'+184)b' + (-93n'+252)c'$$
$$+ 22(2\beta'+3\gamma'+3i') + 27(4\beta'+\gamma'+i')$$
$$+ i' + \beta' - 24C' - 28B' - \iota' - 2\tfrac{1}{3}j' - 3\tfrac{1}{3}\chi' + \theta'\beta' - f'\omega'$$
$$+ 4C + 10B + \iota + 7j + 8\chi - 4\omega. \tag{29}$$

The foregoing equation (26) in fact expresses that the surface and its reciprocal have the same deficiency; viz. the expression for the deficiency is

$$\begin{aligned}\text{Deficiency} &= \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \tfrac{1}{2}(q+r) + 2t + i + \beta + \gamma + i - \theta,\\ &= \tfrac{1}{6}(n'-1)(n'-2)(n'-3) - \&\text{c.} \quad (30)\end{aligned}$$

629. The equation (28) (due to Prof. Cayley) is the correct form of an expression for β', first obtained by him (with some errors in the numerical coefficients) from independent considerations, but which is best obtained by means of the equation (26); and (27) is a relation presenting itself in the investigation. In fact, considering a as standing for its value $n(n-1)-2b-3c$, we have from the first 25 equations

$$\begin{aligned} & 0 = 2(n-1)(n-2)(n-3) - 12(n-3)(b+c) + \dots\\ -2\sigma(n-2):\quad & \kappa - B - \rho - 2\sigma - 3\omega = 0,\\ -4b(n-2):\quad & -\rho - 2\beta - 3\gamma - 3i = 0,\\ -6c(n-2):\quad & -2\sigma - 4\beta - \gamma - i - \omega = 0,\\ +2:\quad & 2q + 2r + 4i + 14\beta + 10\gamma + \dots\end{aligned}$$

and multiplying these equations by the numbers set opposite to them respectively, and adding, we find

$$\begin{aligned}&-2n^3 + 12n^2 + 4n + b(12n-36) + c(12n-48)\\ &\qquad - 6q - 6r - 4i - 10\beta - 4\gamma - 24i - 7j - 8\chi + 2\theta - 4\omega = \Sigma,\end{aligned}$$

and adding thereto (26) we have the equation (27); and from this (28), or by a like process, (29), is obtained without much difficulty. As to the 8 Σ-equations or symmetries, observe that the first, third, fourth, and fifth are in fact included among the original equations (for an expression which vanishes is in fact $=\Sigma$); we have from them moreover $3n - c = 3a' - \kappa'$, and thence $3n - c - \kappa = 3a' - \kappa' - \kappa$, which is $=\Sigma$, or we have thus the second equation; but the sixth, seventh, and eighth equations have yet to be obtained.

630. The equations (15), (16), (17) give

$$\begin{aligned}
n' &= a(a-1) - 2\delta' - 3\kappa', \\
c' &= 3a(a-2) - 6\delta' - 8\kappa', \\
\delta' &= \tfrac{1}{2}a(a-2)(a^2-9) - (a^2-a-6)(2\delta'+3\kappa') + 2\delta'(\delta'-1) + 6\delta'\kappa' + \tfrac{9}{2}\kappa'(\kappa'-1);
\end{aligned}$$

from (7), (8), (9) we have

$$(a-b-c)(n-2) = \kappa - B - 6\beta - 4\gamma - 3i - \theta + 2\omega,$$
$$(a-2b-3c)(n-2)(n-3) = 2(\delta - C) - 8k - 18h - 6c + 18\beta + 12\gamma + 9i - 6\omega),$$

and substituting these values for κ and δ, and for a its value $= n(n-1) - 2b - 3c$, we obtain the values of n', c', b'; viz. the value of n' is

$$\begin{aligned}
n' = n(n-1)^2 - n(7b+12c) + 4b^2 + 8b + 9c^2 + 15c \\
- 2k - 18f + 18\beta + 12\gamma + 12t - 9i - 2\theta - 3j - 3\chi.
\end{aligned}$$

Observe that the effect of a cnicnode C is to reduce the class by 2, and that of a binode B to reduce it by 3.

631. We have

$$(n-2)(n-3) = n^2 - n + (-4n+6) = a + 2b + 3c + (-4n+6),$$

and making this substitution in the equations (10), (11), (12), which contain $(n-2)(n-3)$, these become

$$\begin{aligned}
a(-4n+6) &= 2(\delta - C) - a^2 - 4\rho - 9\sigma - 2j - 3\chi - 1\omega, \\
b(-4n+6) &= 4k - 2b^2 - 9\beta - 6\gamma - 3i - 2\rho - j, \\
c(-4n+6) &= 6h - 3c^2 - 6\beta - 4\gamma - 2i - 3\sigma - \chi + \omega,
\end{aligned}$$

(the foregoing equations (C) Salmon p. 586); and adding to each equation four times the corresponding equation with the factor $(n-2)$, these become

$$\begin{aligned}
a^2 - 2a &= 2(\delta - C) + 4(\kappa - B) - \rho - 2j - 3\chi - 3\omega, \\
2b^2 - 2b &= 4k - \rho + 6\gamma + 12t - 3i + 2\rho - j, \\
3c^2 - 2c &= 6h + 10\beta + 4\rho - 2i + 5\sigma - \chi + \omega.
\end{aligned}$$

Writing in the first of these $a^2-2a=n'+2\delta'+3\kappa'-a$, and reducing the other two by means of the values of q, r, the equations become

$$\begin{aligned}\sigma' &= a^2-2C-4\rho+\kappa-\sigma-2j-3\chi-3\omega,\\ 2q+\beta+3i+j &= 2\rho,\\ 3r+c+2i+\chi &= 5\sigma+\beta+4\rho+\omega,\end{aligned}$$

which give at once the last three of the 8 Σ-equations.

The reciprocal of the first of these is

$$\sigma'=a-n+\kappa'-2j'-3\chi'-2C'-4B'-3\omega',$$

or writing herein $a=n(n-1)-2b-3c$ and $\kappa'=3n(n-2)-6b-8c$, this is

$$\sigma'=4n(n-2)-8b-11c-2j-3\chi'-2C'-4B'-3\omega',$$

giving the order of the spinode curve; viz. for a surface of the order n without singularities this is $=4n(n-2)$, the product of the orders of the surface and its Hessian.

632. Instead of obtaining the second and third equations as above, we may to the value of $b(-4n+6)$ add twice the value of $b(n-2)$; and to twice the value of $c(-4n+6)$ add three times the value of $c(n-2)$, thus obtaining equations free from ρ and σ respectively; these equations are

$$\begin{aligned}b(-2n+2) &= 4k-2b'-5\beta-3i+6t-j,\\ c(-5n+6) &= 12h-6c'-5\gamma-4i-2\chi+3\sigma-3\omega,\end{aligned}$$

equations which, introducing therein the values of q and r, may also be written

$$\begin{aligned}b(2n-4) &= 2q+5\beta+6\gamma+6t+3i+j+4f,\\ c(5n-12)+3\omega &= 6r+18\beta+5\gamma+4i+2\chi+3\omega.\end{aligned}$$

Considering as given, n the order of the surface; the nodal curve with its singularities b, k, f, t; the cuspidal curve and its singularities c, h; and the quantities β, γ, i which relate to the intersections of the nodal and cuspidal curves; the first of the two equations gives j, the number of pinch-points, being singularities of the nodal curve *quoad* the surface; and the second equation establishes a relation between θ, χ, ω, the numbers of singular points of the cuspidal curve *quoad* the surface.

In the case of a nodal curve only, if this be a complete intersection $P=0$, $Q=0$, the equation of the surface is $(A,B,C\mid P,Q)^2=0$, and the first equation is

$$b(-2n+2)=4k-2b^2+6t-j;$$

or, assuming $t=0$, say $j=2(n-1)b-2b^2+4k$, which may be verified; and so in the case of a cuspidal curve only, when this is a complete intersection $P=0$, $Q=0$, the equation of the surface is $(A,B,C\mid P,Q)^2=0$, where $AC-B^2=MP+NQ$; and the second equation is

$$c(-5n+6)=12h-6c^2-2\chi+3\theta-3\omega,$$

or, say

$$2\chi+3\omega=(5n-6)c-6c^2+12h+3\theta,$$

which may also be verified.

633. We may in the first instance out of the 46 quantities consider as given the 14 quantities

$$n; \quad b, k, f, t; \quad c, h, \theta, \chi; \quad \beta, \gamma, i; \quad C, B,$$

then of the 26 relations, 17 determine the 17 quantities

$$a, \delta, \kappa, \rho, \sigma; \quad j, q; \quad r, \omega; \quad n', a', \delta', \kappa'; \quad \rho', \sigma', c', \iota,$$

and there remain the 9 equations (18), (19), (20), (21), (22), (23), (24), (25), (28), connecting the 15 quantities

$$\rho', \sigma'; \quad k', f', t', j', c', h', \theta', \chi', \omega', r'; \quad \beta', \gamma'; \quad C', B'.$$

634. The formulae of course give a, δ, κ, q, ρ, σ in terms of n, b, c, and the other quantities respectively; but they do not directly give the curve-singularities k', f', t', j' of the node-couple torse, nor the curve-singularities h', θ', χ', ω' of the spinode torse, nor do they give β', γ', i', C', B': and until these can be obtained, the theory of reciprocal surfaces must be considered as incomplete.

635. The question of singularities has been considered under a more general point of view by Zeuthen, in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. pp. 1—20, 1871. He attributes to the surface:

A number of singular points, viz. points at any one of which the tangents form a cone of the order μ, and class ν, with $y+\eta$ double lines, of which y are tangents to branches of the nodal curve through the point, and $z+\zeta$ stationary lines, whereof z are tangents to branches of the cuspidal curve through the point, and with u double planes and v stationary planes; moreover, these points have only the properties which are the most general in the case of a surface regarded as a locus of points; and Σ denotes a sum extending to all such points. (The foregoing general definition includes the cnicnodes ($\mu=\nu=2$, $y=\eta=z=\zeta=u=v=0$); and [also, but not properly] the binodes ($\mu=2$, $\nu=1$, $y=\eta=\zeta=$ &c. $=0$), [it includes also the off-points ($\mu=\nu=3$, $z=\zeta=1$, $y=\eta=\xi=0$)].)

And, further, a number of singular planes, viz. planes any one of which touches along a curve of the class μ' and order ν', with $y'+\eta'$ double tangents, of which y' are generating lines of the node-couple torse, $z'+\zeta'$ stationary tangents, of which z' are generating lines of the spinode torse, u' double points and v' cusps; it is, moreover, supposed that these planes have only the properties which are the most general in the case of a surface regarded as an envelope of its tangent planes; and Σ' denotes a sum extending to all such planes. [The definition includes the cnictropes ($\mu'=\nu'=2$, $y'=\eta'=z'=\zeta'=u'=v'=0$), and [also, but not properly] the bitropes ($\mu'=2$, $\nu'=1$, $\eta'=\zeta'=$ &c. $=0$), [it includes also the off-planes ($\mu'=\nu'=3$, $z'=\zeta'=1$, $y'=\eta'=\xi'=0$)].)

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Notes and References.

384. The conclusion arrived at Nos. 27—30 that the transformed curve of the order $D+1$ depends upon $4D-6$ parameters is at variance with Riemann's theorem according to which the number of parameters is $3p-3$, (p Riemann $= D$ Cayley), $= 3D-3$, and this last is the correct value. My erroneous conclusion is referred to in the preface to Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866),

> "Unter den von Riemann behandelten Theilen der Theorie haben wir die Frage nach der Anzahl der Moduln einer Klasse von Abel'schen Functionen ausschliessen zu müssen geglaubt. Diese Frage ist durch die scharfsinnigen Betrachtungen des Herrn Cayley Gegenstand der Controverse geworden: sie ist überhaupt wohl zunächst nur durch tiefe algebraische Untersuchungen endgültig zu entscheiden, für deren Schwierigkeiten die gegenwärtig bekannten Methoden nicht mehr auszureichen scheinen."

In the case D (or p) $= 3$, my value is 10, Riemann's is 9: that the latter is correct was shown by a direct proof in the paper Brill, "Note bezüglich der Zahl der Moduln einer Klasse von algebraischen Gleichungen," *Math. Ann.*, t. i. (1869), pp. 401—406: the explanation of my error is given in the paper, Cayley, "Note on the Theory of Invariants," *Math. Ann.*, t. III. (1871), pp. 268—271.

400. The question here considered, viz., the expression of a binary sextic f in the form $v^3 - u^2$, v and u a cubic and a quadric respectively, forms the basis of the very interesting investigations contained in the Memoir, Clebsch "Zur Theorie der binären Formen sechster Ordnung und zur Dreitheilung der hyperelliptischen Functionen," *Gott. Abh.*, t. xiv. (1869), pp. 1 &c. Considering f as a given sextic it is remarked that the number of solutions, or what is the same thing the number of the functions u or v, although at first sight $= 45$, is really $= 40$; supposing that there is a given solution u, v, or that the sextic function is in the first instance given in the form $v^3 - u^2$, then if any other solution is u', v', we have $v'^3 - v^3 = u'^2 - u^2$, where v', u' are functions to be determined: there are in all 39 solutions, a set of 27 and a set of 12 solutions: viz. writing the equation in the form $(v+v')(v-v') = (u-u')(u-\epsilon u')(u-\epsilon^2 u')$, ϵ an imaginary cube root of unity, then either the $v+v'$ and the $v-v'$ contain each of them as a factor one of the quadric functions $u-u'$, $u-\epsilon u'$, $u-\epsilon^2 u'$ (which gives the set of 27 solutions) or else the $v+v'$ and the $v-v'$ are each of them the product of three linear factors of the quadric functions respectively (which gives the set of 12 solutions). It may be added that the 27 solutions form 9 groups of 3 each and that these 9 groups depend upon Hesse's equation of the order 9 for the determination of the inflexions of a cubic curve; and that the 12 solutions are determined by an equation of the order 12 which is the known resolvent of this order arising from Hesse's equation and is solved by means of a quartic equation with a quadrinvariant $= 0$. As appears by the title of the memoir, the question is connected with that of the trisection of the hyperelliptic functions.

401, 403. On the subject of Pascal's theorem, see Veronese, "Nuove teoremi sull' hexagrammum mysticum," *R. Accad. dei Lincei* (1876—77), pp. 7—61; Miss Christine Ladd (Mrs Franklin), "The Pascal Hexagram," *Amer. Math. Jour.*, t. ii. (1879), pp. 1—12, and Veronese, "Interpretations géométriques de la théorie des substitutions de n lettres, particulièrement pour $n = 3$, 4, 5, en relation avec les groupes de l'Hexagramme Mystique," *Ann. di Matem.*, t. xi. 1882—83, pp. 93—236. See also Richmond, "A Symmetrical System of Equations of the Lines on a Cubic Surface which has a Conical Point," *Quart. Math. Jour.*, t. xxii. (1889), pp. 170—179, where the author discusses a perfectly symmetrical system of the lines on the cubic surface and deduces from them equations of the lines relating to a Pascal's hexagon; there are of course through the conical point 6 lines lying on a quadric cone and these by their intersections with the plane give the six points of the hexagon; the interest of the paper consists as well in the connexion established between the two theories as in the perfectly symmetrical form given to the equations.

406, 407. A correction was made by Halphen to the fundamental theorem of Chasles that the number of the conics (X, R) is $= a\mu + b\nu$, he finds that a diminution is in some cases required, and thus that the general form is, Number of conics $(X, R) = a\mu + b\nu - r$: see Halphen's two Notes, *Comptes Rendus*, 4 Sep. and 13 Nov., 1876, t. LXXXIII. pp. 537 and 886, and his papers "Sur la théorie des caractéristiques pour les coniques," *Proc. Lond. Math. Soc.*, t. IX. (1877—1878), pp. 149—170, and "Sur les nombres des coniques qui dans un plan satisfont à cinq conditions projectives et indépendantes entre elles," *Proc. Lond. Math. Soc.*, t. X. (1878—79), pp. 76—87: also Zeuthen's paper "Sur la revision de la théorie des caractéristiques de M. Study," *Math. Ann.*, t. XXXVII. (1890), pp. 461—464, where the point is brought out very clearly and tersely.

The correction rests upon a more complete development of the notion of the line-pair-point, viz. this degenerate form of conic seems at first sight to depend upon three parameters only, the two parameters which determine the position of the coincident lines, and a third parameter which determines the position therein of the coincident points: but there is really a fourth parameter. [Compare herewith the point-pair, or indefinitely thin conic, which working with point-coordinates presents itself in the first instance as a coincident line-pair depending on two parameters only, but which really depends also on the two parameters which determine the position therein of the vertices.]

As to the fourth parameter of the line-pair-point the most simple definition is a metrical one; taking the semiaxes of the degenerate conic to be a and b ($a = 0$, $b = 0$) then we have two positive integers p and q prime to each other such that the ratio $a^q : b^p$ is finite; and this being so the fractional or it may be integer number $p : q$ is the fourth parameter in question. But it is preferable to adopt Halphen's purely descriptive definition, viz. we consider a conic 1° in reference to three given points y, z, t on a given line, and take x, x' for the remaining intersections of the conic with the lines yzt, ytz respectively; the fourth parameter is the anharmonic ratio of the pencil x, x', y, z.

408, 409. As to the correction, see Halphen, "Sur une série de courbes analogues aux courbes de Lamé," *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

411. The "Addition" referred to in the text is given as [415] of the present volume.

412. For the curve $x^3 + y^3 + z^3 + 6lxyz = 0$, the sextactic points are given as the intersections of the curve by the Hessian $l(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0$. But these are the points of contact of the tangents from the 9 flexions; in fact the equation of a tangent is $x(\alpha^2 + 2l\beta\gamma) + \ldots = 0$, and for the point of contact we have $x : y : z = \alpha(\alpha^3 + 2l\beta\gamma) : \ldots$; substituting in the equation of the curve, this becomes $(\alpha^3+\beta^3+\gamma^3+6l\alpha\beta\gamma)(\alpha^6+\ldots-10l\alpha^3\beta^3\gamma^3+2l^2\beta^3\gamma^3\cdot\alpha^3+\ldots) = 0$; the first factor vanishes for a point on the curve, and the second factor is $= (\alpha^3 + \beta^3 + \gamma^3)^2 - 4(1 + 2l^3)^2\alpha^3\beta^3\gamma^3$ which $= 0$ gives the 9 points of contact. But these are the intersections of the curve by the Hessian; and hence the theorem.

413. The theorem that a cubic curve has 27 sextactic points was first given in my paper "Note on the Inflections of a Cubic Curve," *Quart. Math. Jour.*, t. XII. (1873), pp. 365—366 [417]; see also the paper "On the Sextactic Points of a Plane Curve," *Phil. Trans.*, vol. CLXV. (for the year 1875), pp. 545—578 [428].

414. The formulae for the class of the envelope of the osculating conics were verified by Mr. T. Cotterill in the particular case of a cubic curve; see his paper "On the Number and Nature of the Sextactic Points of a Plane Curve," *Quart. Math. Jour.*, t. XII. (1873), pp. 132—143.

415. The corrections and additions here made are given in a more complete form in [416].

416. The formulae given in this Addition require some explanation in reference to the singularities χ, χ' and ω, ω' which present themselves. In Salmon's original formulae (A), (B), the terms in χ were omitted; the terms in ω were subsequently introduced by Dr Zeuthen. The singularities in question are:

χ, close-points: these are points on the cuspidal curve, not ordinary points thereof, but points where the cuspidal curve has a certain singularity analogous to that of the point of contact of a stationary tangent; or say, they are points where the cuspidal curve is met by its envelope. The number of them is χ.

ω, off-points: these are isolated singular points of the surface, not situate on either the nodal or the cuspidal curve.

And reciprocally:

χ', close-planes; ω', off-planes.

417. In the present state of the theory of surfaces, the only case in which we can be certain of the complete system of singularities is that of the general surface of the order n without any singularities; for this case $n' = n(n-1)^2$.

418. The expression for the deficiency given in the text is not universally applicable; the deficiency of a surface is not in general determined by the order of the surface and the orders of its nodal and cuspidal curves only, but depends also on the manner in which these curves are situate upon the surface. The question is considered by Zeuthen, "Note sur la déficience des surfaces algébriques," *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

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419. The theory of reciprocal surfaces is considered under a more general point of view by Zeuthen in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. (1871), pp. 1—30; and also in the memoir "Sur la déficience des surfaces algébriques," *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

420. With Zeuthen, the nodal curve has

$$3t + f + 3\theta' + \chi' \quad \text{double points}$$

($k = i + 3t + f + 3\theta' + 2\chi'$, if k denotes, as with me, the number of apparent double points of the curve), and it has

$$\gamma + d + 3\chi' + \xi' \quad \text{stationary points.}$$

The cuspidal curve has

$$g + 6\chi' + 12\xi' + U' + 4\theta' + 2\Sigma + 2\Sigma' \quad \text{double points}$$

($h = h + g + 6\chi' + 12\xi' + U' + 4\theta' + 2\Sigma + 2\Sigma'$, if h denotes, as with me, the number of apparent double points of the curve), and it has

$$\beta + 6 + 2\theta' + \Sigma \quad \text{stationary points}$$

and the nodal and cuspidal curves intersect in

$$3\beta + 2\gamma + i + 12\theta' + 2\Sigma + 2\Sigma' \quad \text{points;}$$

where I have written Σ and Σ' to denote sums (different in the different equations) determined by Zeuthen, and depending on the singularities K and S' respectively.

For comparison of my formulae with Zeuthen's it is thus proper in my formulae to write $\theta = 0$, $\omega = 0$, $\chi = 0$ (but in the first instance I retain θ) and in his formulae to write $\theta' = 0$, $\chi' = 0$, $d = 0$, $g = 0$, $e = 0$, $\Sigma = 0$, $\Sigma' = 0$. Doing this the last mentioned formulae give as with me $3t+f$ double points and γ stationary points for the nodal curve, but they give for the cuspidal curve $6\chi' + 12\xi'$ (instead of 0) double points and β stationary points; and the two curves intersect (as with me) in $3\beta + 2\gamma + i$ points.

There is a real discrepancy in the number $6\chi' + 12\xi'$ of double points on the cuspidal curve.

I compare his $(6+2\times 6+1=33)$ relations

$$\begin{array}{ll} a=a', & d=d', \\ i=i', & g=g', \\ \kappa=\kappa', & h=h', \end{array}$$

$$n(n-1)=a+2b+3c. \tag{6}$$

$$a(a-1)=n'+2\delta'+3\kappa'. \tag{7}$$

$$c-\kappa'=3(n-a). \tag{8}$$

$$b(b-1)=g+2k+3j+3\omega'+\chi'. \tag{9}$$

$$\tfrac{1}{2}(b-q)=\gamma+d-s+\Sigma', \quad [\text{determines } s]. \tag{10}$$

$$c(c-1)=r+2h+3\beta+6\theta'+3e. \tag{11}$$

$$\tfrac{1}{2}(c-r)=\beta+e-m+2\theta'+2\Sigma, \quad [\text{determines } m]. \tag{12}$$

$$a(n-2)=\kappa-\beta+\rho+2\sigma+\Sigma. \tag{13}$$

$$b(n-2)=\rho+2\beta+3\gamma+3i+9\theta'+\chi'. \tag{14}$$

$$c(n-2)=2\sigma+4\beta+\gamma+8\chi'+16\xi'+12\theta'+\Sigma. \tag{15}$$

$$a(n-2)(n-3)=2(\delta-3C)+3(ac-3\sigma-\chi)+2(ab-2\rho-j). \tag{16}$$

$$b(n-2)(n-3)=4(k-3\xi')+(ab-2\rho-j)+3(bc-3\beta-2\gamma-i)+3\chi'+2\Sigma'. \tag{17}$$

$$\begin{aligned} c(n-2)(n-3)=\;&6(h-6\chi'-12\xi'-U'-4\theta'-\Sigma)\\ &+(ac-3\sigma-\chi)+2(bc-3\beta-2\gamma-i)\\ &-30\theta'+2\Sigma+2\Sigma'. \end{aligned} \tag{18}$$

with the like reciprocal equations $(6')$ to $(18')$

$$\begin{aligned} &a+m-r-\theta-\chi'-4\eta'+\Sigma'=0,\\ &\qquad 4\mu'-3\xi'-14\eta'+\Sigma=\theta'+m'-f'-\beta'-4j-3\chi-14\eta+\Sigma', \end{aligned}$$

—where κ, &c.

and my $(3+22+1=26)$ relations as follows:

$$a=a', \qquad\qquad f=f'. \tag{1}$$

$$i=i'. \tag{2}$$

$$\kappa=\kappa'. \tag{3}$$

$$a=n(n-1)-2b-3c. \tag{4}$$

$$\kappa'=3n(n-2)-6b-8c. \tag{5}$$

$$\delta'=\tfrac{1}{2}n(n-2)(n^2-9)-\&\text{c}. \tag{6}$$

$$\text{(A)}\quad q=b'-b-2k-2f-3j-6t. \tag{13}$$

$$\text{(B)}\quad r=c'-c-2h-3\beta. \tag{14}$$

$$\text{(C)}\quad a(n-2)=\kappa-B+\rho+2\sigma+3\omega. \tag{7}$$

$$\text{(D)}\quad b(n-2)=\rho+2\beta+3\gamma+3i. \tag{8}$$

$$\text{(E)}\quad c(n-2)=2\sigma+4\beta+\gamma+i+\omega. \tag{9}$$

$$\text{(F)}\quad a(n-2)(n-3)=2(\delta-C-3\omega)+3(ac-3\sigma-\chi-3\omega)+2(ab-2\rho-j). \tag{10}$$

$$\text{(G)}\quad b(n-2)(n-3)=4k+(ab-2\rho-j)+3(bc-3\beta-2\gamma-i). \tag{11}$$

$$\text{(H)}\quad c(n-2)(n-3)=6h+(ac-3\sigma-\chi-3\omega)+2(bc-3\beta-2\gamma-i), \tag{12}$$

with the like reciprocal equations $(4')$ to $(14')$;

$$\begin{aligned}\text{(I)}\quad & 2(n-1)(n-2)(n-3)-12(n-3)(b+c)+6q+6r+24i+42\beta+30\gamma-\chi \\ &=2(n'-1)(n'-2)(n'-3)-12(n'-3)(b'+c')+6q'+6r'+24i'+42\beta'+30\gamma'-\chi'.\end{aligned} \tag{26}$$

$$\begin{gathered}k+\xi+3t+3\theta'+\chi'-E,\\ h+\theta'+6\chi'+12\xi'+U'+4\theta'+\Sigma+2\Sigma'.\end{gathered}$$

Substituting for κ; h their values we have instead of (A), (B), (C), (D) the equations

$$\begin{aligned}
&\text{(A')}\quad b' - b = q + 2k + 2f + 6t + 6\theta' + 3j + 3\chi' + \Sigma + \Sigma',\\
&\text{(B')}\quad c' - c = r + 2h + 2g + 12\chi' + 2U' + 2\xi' + W' + 3\theta' + 3e + \Sigma + \Sigma',\\
&\text{(C')}\quad b(n-2)(n-3) = 4k + 2\chi'\theta' + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i) + \Sigma + \Sigma',\\
&\text{(D')}\quad c(n-2)(n-3) = 6h - 30\theta' + (ac - 3\sigma - \chi) + 2(bc - 3\beta - 2\gamma - i) + \Sigma + \Sigma'.
\end{aligned}$$

Writing as before $C = 0$, $\omega = 0$; $U = 0$, $\chi' = 0$, $d = 0$, $g = 0$, $e = 0$, and neglecting the terms in Σ, Σ', the two equations (E) become

$$\begin{aligned}
&\text{Zeuthen}\quad c(n-2) = 2\sigma + 4\beta + \gamma + 8\chi' + 16\xi',\\
&\text{Cayley}\quad c(n-2) = 2\sigma + 4\beta + \gamma + \theta,
\end{aligned}$$

which can be made to agree by writing $\theta = 8\chi' + 16\xi'$. But we have

$$\begin{aligned}
&\text{Zeuthen}\quad \text{(F)}\quad c' - c = r + 2h + 3\beta + 12\chi' + 24\xi',\\
&\text{Cayley}\quad (5)\quad c' - c = r + 2h + 3\beta,
\end{aligned}$$

values which differ by the terms $12\chi' + 24\xi'$, or if θ has the value just written down, the term $\frac{1}{2}\beta$.

I refrain from a comparison of the two equations (I), — and of the expressions for the deficiency given by these two equations respectively; but I notice here the expression for the deficiency obtained by Zeuthen in the last section (XIV.) of his Memoir, viz. this is

$$\begin{aligned}
24(D+1) = c' - 12n + 24a + \Sigma\delta + 3\kappa - 15c + 2\sigma + 6\rho + 12\chi' + 6g + 9e&\\
+ 8B + 2U' + 18\theta + 6U' + 6\theta'&\\
+ \Sigma(3\kappa' + 3\xi' + 8\xi + 13\eta) + \Sigma'(6\eta')&.
\end{aligned}$$

The problem is a very difficult one, and it cannot be held that as yet a complete solution has been obtained. Take in plane geometry the question of reciprocal curves; here, using throughout point-coordinates, we start with a curve represented by the general equation $(x, y, z)^n = 0$, such a curve has only isolated singularities, viz. the line-singularities of the inflexion and the double tangent, we know the expression in point-coordinates of any such singularity (inflexion or double tangent as the case may be), viz. we can at once write down the equation of a curve of the order n having a given stationary tangent and point of contact therewith, or a given double tangent and two points of contact therewith. Returning to the general curve $(x, y, z)^n = 0$, we know that the reciprocal curve has other isolated singularities, viz. the point-singularities which correspond to these, the double point (or node) and the stationary point (or cusp), and we know the expression of any such singularity (node or cusp as the case may be), viz. we can at once write down the equation of a curve of the order n having at a given point a node with given tangents, or a cusp with given tangent. And then starting afresh with a curve of the order n having a node or a cusp we obtain the effect thereof as regards the line-singularities of the inflexion and the double tangent. We are thus led to consider as ordinary singularities in the theory the above-mentioned four singularities of the inflexion, the double tangent, the node and the cusp: and we know further that any other singularity whatever of a plane curve is compounded in a definite manner of a certain number of some or all of these singularities.

But in the theory of surfaces, starting in like manner with the general equation $(x, y, z, w)^n = 0$, such a surface has torse-singularities, the node-couple torse, and the spinode-torse; each of these is in general an indecomposable torse of a certain kind (but there is the new cause of complication that it may break into two or more separate torses), but we do not know the analytical expression of these singularities, nor consequently the analytical expression of the curve-singularities which correspond to them, the nodal curve and the cuspidal curve. Thus if we attempt to start with a surface $(x, y, z, w)^n = 0$ having a nodal curve, we can indeed write down the equation in its most general form, viz. if the nodal curve has for its complete expression the k equations $P = 0$, $Q = 0$, $R = 0$, &c. (viz. if the curve is such that every surface whatever through the curve is of the form $\Omega = AP+BQ+CR+\ldots = 0$) then the most general equation of the surface having this curve for a nodal curve is $(A, B, C, \ldots \mid P, Q, R, \ldots)^2 = 0$, but this form is far too complicated to be worked with; and if for simplicity we take the nodal curve to be a complete intersection $P = 0$, $Q = 0$, and consequently the equation of the surface to be $(A, B, C \mid P, Q)^2 = 0$, then it is by no means clear that we do not in this way introduce limitations extraneous to the general theory. The same difficulty applies of course, and with yet greater force, to the cuspidal curve; and even if we could deal separately with the cases of a surface having a given nodal curve, and a given cuspidal curve, this would in no wise solve the problem for the more general case of a surface having a given nodal curve and a given cuspidal curve. It is to be added that the general surface of the order n has no plane- or point-singularities, and thus that such singularities (which correspond most nearly to the singularities considered in the theory of reciprocal curves) present themselves in the theory of reciprocal surfaces as extraordinary singularities.

End of Vol. VI.

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